Invariants

Invariant: a property. held by a class of mathematical objects, which remains uncharged when transformations of a certain type are applied to the object.

Problem 1

1 is written on the board.
Every minute, Titan adds 2 to the number.
Is it possible that 2022 will be written?

→ NO because the parity of the number does not change.
It is INVARIANT.

Problem 2

1 is written on the board.
Every minute, Titan adds the sum of the digits to the number. Is it possible to get 2022?

$$1 \to 2 \to 4 \to 8 \to 16 \to 23 \to 28 \to 38$$

(mod 3) 1 2 1 2 1 2 1 2

Solution: This is impossible.
We know that $X \underset{3}{\equiv}$ Sum of the digits of X.
Hence if $y = X + S(X)$
then $y \pmod 3 \equiv X + X = 2X$
→ If X is not a multiple of 3, then y isn't either.
But $3 \mid 2022$. QED.
The non-divisibility by 3 is the invariant.

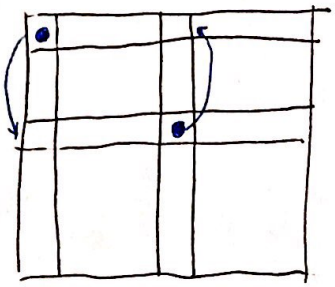
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Look at HW problem:

Can you put 8 rooks on a chessboard such that no two attack each other and exactly 3 rooks are on white squares?

Examine 2 rooks that don't attack each other.

Keeping each in their original columns, exchange their rows. That way, you keep those two rows and columns occupied by one rook each.



- If one rook moves to a square of the same color as its original square, the other one will also.
- If one rook moves to a square of a different color from its original, the other one will also.

The parity of the number of rooks on white squares does not change!

Since we know the diagonal of rooks on 8 black squares or 8 white squares is one configuration where no 2 attack each other, the number of rooks on white squares must stay even.

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